

# Lucinda (Luca) Higgins

San Francisco Bay Area, CA | [latwist@stanford.edu](mailto:latwist@stanford.edu) | <https://www.linkedin.com/in/lucahiggins/>

---

# EDUCATION

**Bachelor of Science in Mechanical Engineering**  
Stanford University, Stanford, CA

*Expected June 2028*

**Relevant Coursework:** Linear Algebra & Differential Equations • Dynamics • Solid Mechanics • Mechanics of Materials • Engineering Thermodynamics • Fluid Mechanics • Heat Transfer • Product Realization • Electricity & Magnetism

---

# EXPERIENCE

**Undergraduate Research Assistant**

*February 2025 - present*

[Soft Intelligent Materials Laboratory](#), PI: Professor Renee Zhao, Dept. of Mechanical Engineering, Stanford, California

Develop magnetically actuated milli-scale robots for minimally invasive thrombectomy through computational fluid dynamics (CFD), rapid prototyping, and experimental validation. Optimized robot geometry to achieve a 5× improvement in swimming speed and presented research on the [Magnetic Milli-Spinner](#) at the Bay Area Robotics Symposium. Contribute to the design of origami-inspired reconfigurable structures for soft robotic applications. See [Optimization of Magnetic Milli-Spinner for Robotic Endovascular Intervention](#), Lu, Higgins, Bernardo, Zhao (2026).

**Mechanical Engineering Team Member**

*February 2026 - present*

**BuilderBot and BattleBots**, [Stanford Student Robotics Club](#), Stanford, California

Design and fabricate modular combat robot subsystems with hot-swappable drive and weapon assemblies for rapid repairs during competition. Develop structural components for a 6-DOF robotic arm using CAD and additive manufacturing while collaborating across mechanical, electrical, and software teams to integrate ROS 2-controlled hardware with AI-based visuomotor control.

**Co - President**

*May 2025 - June 2026*

[Stanford Student Robotics Club](#), Stanford, California

Led Stanford Student Robotics, a 150+ member engineering organization supporting competitive teams, research initiatives, and educational programs. Directed a nine-member executive board, managed an annual operating budget exceeding \$100,000, and established partnerships with the Stanford Robotics Center, alumni, industry sponsors, and university leadership.

**Mechanical Engineering Team Member**

*September 2024 - February 2026*

[VTOL Drone \(Skyrunners\)](#), Stanford Student Robotics Club, Stanford, California

Led mechanical engineering for an autonomous fixed-wing drone delivery platform. Designed and manufactured custom mechanical components through iterative prototyping and additive manufacturing while contributing to flight testing, systems integration, and presentation of the Mission Concept Review.

**Mechanical Engineering Team Member**  
2026

*September 2024 - March*

[Robosub Competition Team](#), Stanford Student Robotics Club, Stanford, California

Designed and manufactured custom mechanical components for an autonomous underwater vehicle competing in RoboSub. Developed CAD models, fabricated components using additive manufacturing, and iterated designs to improve reliability in underwater environments.

**Financial Officer**

*February 2025 - May 2025*

[Stanford Student Robotics Club](#), Stanford, California

Managed financial operations for a 150+ member engineering organization supporting five technical teams. Oversaw budgeting, reimbursements, sponsorships, and long-term financial planning.

---

# TECHNICAL SKILLS

CAD: SolidWorks, Fusion 360(FEA), Onshape, Simulation: COMSOL (CFD), MATLAB, Manufacturing: Manual Mill, Lathe, Laser Cutting, 3D Printing, Sand Casting, Programming: Java, MATLAB, Languages: English (Fluent), Mandarin Chinese (Intermediate)